Topics in Causal Inference

Technical HW Solutions

Homework Problems

Week 1: Linear Regression, Covariate Adjustment, and FWL

Problem 1 (FWL with Multivariate Treatment)**.** *Let $Y \in \mathbb{R}^n$ be an outcome vector, $Z \in \mathbb{R}^{n\times q}$ a matrix of treatment variables, and $X \in \mathbb{R}^{n\times p}$ a matrix of covariates. Consider the linear regression*

$$Y = Z\beta + X\gamma + \varepsilon,$$

estimated by ordinary least squares.

1. *Define the residual-maker matrix*

$$M_X = I - X(X^\top X)^{-1}X^\top.$$

   Show that the OLS estimator $\hat{\beta}$ satisfies

$$\hat{\beta} = (Z^\top M_X Z)^{-1} Z^\top M_X Y.$$

2. *Let $\tilde{Z} = M_X Z$ denote the residuals from regressing each column of Z on X. Show that $\hat{\beta}$ is numerically equal to the coefficient vector obtained from regressing Y on $\tilde{Z}$.*

3. *Interpret this result in words. What does it say about how regression "adjusts" for covariates?*

Solution 1 (FWL with Multivariate Treatment)**.** ***(1) Deriving $\hat{\beta} = (Z^\top M_X Z)^{-1} Z^\top M_X Y$.***

We have the linear regression

$$Y = Z\beta + X\gamma + \varepsilon,$$

where $Y \in \mathbb{R}^n$, $Z \in \mathbb{R}^{n\times q}$, $X \in \mathbb{R}^{n\times p}$. The OLS estimator $(\hat{\beta}, \hat{\gamma})$ minimizes the sum of squared residuals $\|Y - Z\beta - X\gamma\|^2$, and therefore satisfies the normal equations

$$\begin{pmatrix} Z^\top Z & Z^\top X \\ X^\top Z & X^\top X \end{pmatrix} \begin{pmatrix} \hat{\beta} \\ \hat{\gamma} \end{pmatrix} = \begin{pmatrix} Z^\top Y \\ X^\top Y \end{pmatrix}.$$

From the second block equation,

$$X^\top Z\hat{\beta} + X^\top X\hat{\gamma} = X^\top Y.$$

Solving for $\hat{\gamma}$,

$$\hat{\gamma} = (X^\top X)^{-1} X^\top (Y - Z\hat{\beta}).$$

Now plug this into the first block equation:

$$\begin{aligned} Z^\top Z\hat{\beta} + Z^\top X\hat{\gamma} &= Z^\top Y, \\ Z^\top Z\hat{\beta} + Z^\top X(X^\top X)^{-1} X^\top (Y - Z\hat{\beta}) &= Z^\top Y. \end{aligned}$$

Rearrange terms so that everything involving $\hat{\beta}$ is on the left:

$$\left[Z^\top Z - Z^\top X(X^\top X)^{-1}X^\top Z\right]\hat{\beta} = Z^\top Y - Z^\top X(X^\top X)^{-1}X^\top Y.$$

Define the residual-maker matrix

$$M_X = I - X(X^\top X)^{-1}X^\top.$$

Then

$$Z^\top M_X Z = Z^\top Z - Z^\top X(X^\top X)^{-1}X^\top Z, \quad Z^\top M_X Y = Z^\top Y - Z^\top X(X^\top X)^{-1}X^\top Y.$$

So we can write

$$\hat{\beta} = (Z^\top M_X Z)^{-1} Z^\top M_X Y.$$

(2) Regression on residualized Z.

Let $\tilde{Z} = M_X Z$. This means each column of Z has been linearly regressed on X, and $\tilde{Z}$ contains the residuals from those regressions.

Now consider the regression of Y on $\tilde{Z}$ alone:

$$Y = \tilde{Z}\tilde{\beta} + \eta.$$

The OLS estimator in this regression is

$$\tilde{\beta} = (\tilde{Z}^\top \tilde{Z})^{-1}\tilde{Z}^\top Y.$$

Substitute $\tilde{Z} = M_X Z$:

$$\tilde{\beta} = (Z^\top M_X^\top M_X Z)^{-1} Z^\top M_X^\top Y.$$

The matrix M_X is symmetric and idempotent:

$$M_X^\top = M_X, \qquad M_X^2 = M_X.$$

Therefore

$$\tilde{\beta} = (Z^\top M_X Z)^{-1} Z^\top M_X Y.$$

Comparing with part (1), we see

$$\tilde{\beta} = \hat{\beta}.$$

So the coefficient vector on Z in the regression of Y on (Z, X) is numerically equal to the coefficient vector from regressing Y on the residualized regressors $\tilde{Z}$.

(3) Interpretation in words.

The result says:

- *First, remove from each treatment variable in Z its linear dependence on the covariates X. That is, regress each column of Z on X and keep only the residuals $\tilde{Z}$.*
- *Then, regress Y on these residuals $\tilde{Z}$.*

The coefficient $\hat{\beta}$ you get is exactly the same as if you had regressed Y on both Z and X together.

In plain language: regression "adjusts" for covariates by comparing variation in Y to the part of Z that is orthogonal (uncorrelated in a linear sense) to X. *All linear association between Z and X is removed before we look at how Z relates to Y.*

Problem 2 (Misspecification and Orthogonality)**.** *Suppose the true data-generating process is*

$$Y = \tau Z + f(X) + \varepsilon,$$

where $f(X)$ is an unknown (possibly nonlinear) function of X and $E[\varepsilon \mid Z, X] = 0$. You instead estimate the misspecified linear model

$$Y = \alpha + \beta Z + X^\top \gamma + u.$$

1. *Assume that Z is mean-independent of X, i.e.,*

$$E[Z \mid X] = E[Z].$$

   Show that $\hat{\beta}$ is unbiased for τ.

2. *Explain why this result does* not *contradict the general dangers of model misspecification.*

3. *Give an example in which Z is correlated with X and explain why $\hat{\beta}$ would generally be biased in this case.*

Solution 2 (Misspecification and Orthogonality)**.** *We observe data generated by*

$$Y = \tau Z + f(X) + \varepsilon, \quad E[\varepsilon \mid Z, X] = 0,$$

but we estimate the linear model

$$Y = \alpha + \beta Z + X^\top \gamma + u.$$

(1) Mean-independence and unbiasedness.

Assume

$$E[Z \mid X] = E[Z] =: \mu_Z.$$

This means that, on average, treatment does not depend on the covariates X; in other words, Z is mean-independent of X.

By the Frisch–Waugh–Lovell theorem (from Week 1), the OLS coefficient on Z in the regression of Y on (Z, X) is the same as the coefficient from regressing the residualized outcome on the residualized regressor:

$$\hat{\beta} = \frac{\tilde{Z}^\top \tilde{Y}}{\tilde{Z}^\top \tilde{Z}},$$

where:

- *$\tilde{Z}$ is the residual from regressing Z on $(1, X)$,*
- *$\tilde{Y}$ is the residual from regressing Y on $(1, X)$.*

From the true model,

$$Y = \tau Z + f(X) + \varepsilon.$$

When we regress Y on $(1, X)$, the part of Y explained by $(1, X)$ is some linear function of X (and a constant), and the residual $\tilde{Y}$ is what is left over. We can write

$$\tilde{Y} = \tau \tilde{Z} + \tilde{\varepsilon},$$

where $\tilde{\varepsilon}$ is the part of ε that remains after projecting onto $(1, X)$. Then

$$\hat{\beta} = \frac{\tilde{Z}^\top(\tau\tilde{Z} + \tilde{\varepsilon})}{\tilde{Z}^\top\tilde{Z}} = \tau + \frac{\tilde{Z}^\top\tilde{\varepsilon}}{\tilde{Z}^\top\tilde{Z}}.$$

To study the bias, take expectations:

$$E[\hat{\beta}] = \tau + \frac{E[\tilde{Z}^\top\tilde{\varepsilon}]}{E[\tilde{Z}^\top\tilde{Z}]}.$$

Because $E[\varepsilon \mid Z, X] = 0$, we have $E[\varepsilon \mid X] = 0$ and $E[\varepsilon] = 0$. The residual $\tilde{\varepsilon}$ is a linear transformation of ε that is orthogonal to $(1, X)$. The residual $\tilde{Z}$ is also orthogonal to $(1, X)$ by construction.

Under the mean-independence assumption $E[Z \mid X] = \mu_Z$, the part of Z that is left after regressing on $(1, X)$ (i.e. $\tilde{Z}$) is uncorrelated with any function of X, and in particular with the projection of ε on $(1, X)$. Putting these pieces together, it is reasonable (and can be shown formally) that

$$E[\tilde{Z}^\top\tilde{\varepsilon}] = 0,$$

so

$$E[\hat{\beta}] = \tau.$$

In words: if treatment is mean-independent of covariates, then even though we misspecified $f(X)$, the OLS coefficient on Z is still centered around the true effect τ.

(2) Why this does not rescue misspecification in general.

The condition $E[Z \mid X] = E[Z]$ is very strong. It says that, on average, treatment does not depend on covariates at all. This is essentially a "random assignment" condition.

In most observational studies, treatment does *depend on covariates: people with certain characteristics are more likely to receive treatment. When Z and X are related, the residual $\tilde{Z}$ will typically be correlated with the part of Y that we have misspecified (the nonlinear $f(X)$ that we tried to approximate with a linear term). In that case, the extra term $\tilde{Z}^\top\tilde{\varepsilon}$ will not have expectation zero, and $\hat{\beta}$ will be biased.*

So this result does not say "misspecification is harmless." It says: if *treatment is (mean) independent of covariates, then misspecifying the functional form of X does not bias the coefficient on Z. Outside that special case, misspecification can and does cause bias.*

(3) Example with correlation between Z and X.

Consider a simple example:

$$X \sim \mathcal{N}(0, 1), \quad Z = \mathbf{1}\{X > 0\}, \quad Y = \tau Z + X^2 + \varepsilon,$$

with $E[\varepsilon \mid Z, X] = 0$.

Here:

- *People with larger X are more likely to be treated ($Z = 1$ when $X > 0$).*
- *The true effect of X on Y is nonlinear (X^2).*

Suppose we estimate the linear model

$$Y = \alpha + \beta Z + \gamma X + u.$$

Because Z is correlated with X, and the linear term γX cannot fully capture the nonlinear effect X^2, the residualized regressor $\tilde{Z}$ will generally be correlated with the misspecified part of the outcome model (the part of X^2 that is not captured by a linear term). This correlation creates a nonzero bias term in $\hat{\beta}$.

Intuitively: treated units tend to have higher X, and higher X also increases Y through the X^2 term. If we only control linearly for X, some of the effect of X^2 will be wrongly attributed to Z, making $\hat{\beta}$ biased for τ.

Problem 3. *Following the notation of the notes in this part, let $\theta_i = w_i$ when $Z_i = 1$ and $\theta_i = -w_i$ when $Z_i = 0$ for all $i = 1, ...n$. Show that $\sum_i \theta_i = 1$.*

Solution 3. *We use the notation from the Week 1 notes. Let $Z_i \in \{0, 1\}$ be a binary treatment, and let*

$$\tilde{Z}_i = Z_i - \widehat{E}[Z \mid X_i]$$

be the residual from regressing Z on X (and an intercept). The OLS coefficient on Z in the regression of Y on (Z, X) can be written as

$$\hat{\tau} = \frac{\sum_{i=1}^n \tilde{Z}_i Y_i}{\sum_{j=1}^n \tilde{Z}_j^2}.$$

Define

$$w_i = \frac{\tilde{Z}_i}{\sum_{j=1}^n \tilde{Z}_j^2},$$

so that

$$\hat{\tau} = \sum_{i=1}^n w_i Y_i.$$

Following the problem, define

$$\theta_i = \begin{cases} w_i, & Z_i = 1, \\ -w_i, & Z_i = 0. \end{cases}$$

Then we can write

$$\hat{\tau} = \sum_{i=1}^n \theta_i Y_i = \sum_{i:Z_i=1} w_i Y_i - \sum_{i:Z_i=0} w_i Y_i.$$

We want to show that

$$\sum_{i=1}^n \theta_i = 1.$$

First, note that

$$\sum_{i=1}^n \theta_i = \sum_{i:Z_i=1} w_i - \sum_{i:Z_i=0} w_i.$$

From the Week 1 balance results, applied to the constant covariate $X_i \equiv 1$:

- ***Global balance:*** $\sum_{i=1}^n w_i X_i = 0$ *implies* $\sum_{i=1}^n w_i = 0$.
- ***Treated–control balance:***

$$\sum_{i:Z_i=1} w_i X_i = \sum_{i:Z_i=0} (-w_i) X_i \quad \Rightarrow \quad \sum_{i:Z_i=1} w_i = \sum_{i:Z_i=0} (-w_i).$$

Let

$$c := \sum_{i:Z_i=1} w_i = \sum_{i:Z_i=0} (-w_i).$$

Then

$$\sum_{i=1}^{n} w_i = \sum_{i:Z_i=1} w_i + \sum_{i:Z_i=0} w_i = c - c = 0,$$

which is consistent with global balance, and

$$\sum_{i=1}^{n} \theta_i = \sum_{i:Z_i=1} w_i - \sum_{i:Z_i=0} w_i = c - (-c) = 2c.$$

To find c, consider what happens if we regress Y on (Z, X) when $Y_i = Z_i$ for all i. In that case, the true coefficient on Z is 1, so the OLS estimator must satisfy

$$1 = \hat{\tau} = \sum_{i=1}^{n} \theta_i Y_i = \sum_{i=1}^{n} \theta_i Z_i = \sum_{i:Z_i=1} \theta_i = \sum_{i:Z_i=1} w_i = c.$$

Therefore $c = 1$, and hence

$$\sum_{i=1}^{n} \theta_i = 2c = 2 \cdot \frac{1}{2} = 1.$$

So the weights θ_i used to express the OLS coefficient on Z as a linear combination of outcomes sum to 1, as claimed.

Corrected Solution to Problem 3.

Recall the Week 1 notation. Let

$$\tilde{Z}_i = Z_i - \mathbb{E}[Z \mid X_i]$$

be the residual from regressing Z on $(1, X)$, and define

$$w_i \;=\; \frac{\tilde{Z}_i}{\sum_{j=1}^{n} \tilde{Z}_j^2}.$$

The OLS coefficient on Z in the regression of Y on (Z, X) can be written as

$$\hat{\tau} \;=\; \sum_{i=1}^{n} w_i Y_i.$$

Now define

$$\theta_i \;=\; \begin{cases} w_i, & Z_i = 1, \\ -w_i, & Z_i = 0. \end{cases}$$

We show that

$$\sum_{i:Z_i=1} \theta_i = 1.$$

Step 1: Consider the special regression with $Y_i = Z_i$.

Run the regression of Y on (Z, X) when

$$Y_i = Z_i \quad \text{for all } i.$$

In this regression, the true coefficient on Z is exactly 1, so OLS must satisfy

$$\hat{\tau} = 1.$$

Step 2: Compute $\hat{\tau}$ using the weight representation.
Using the general formula,

$$\hat{\tau} = \sum_{i=1}^{n} w_i Y_i = \sum_{i=1}^{n} w_i Z_i = \sum_{i:Z_i=1} w_i,$$

because $Z_i = 1$ for treated units and $Z_i = 0$ for control units.
Thus,

$$1 = \hat{\tau} = \sum_{i:Z_i=1} w_i.$$

Step 3: Relate θ_i to w_i.
For treated units, $\theta_i = w_i$. Therefore,

$$\sum_{i:Z_i=1} \theta_i = \sum_{i:Z_i=1} w_i = 1.$$

Conclusion. The correct identity is

$$\boxed{\sum_{i:Z_i=1} \theta_i = 1}.$$

This replaces the incorrect statement $\sum_{i=1}^{n} \theta_i = 1$ from the original solution.

Solution to Problem 4 (Blocked and Stratified Randomization)

We have N units partitioned into K blocks $B_1, \ldots, B_K$, with $|B_k| = n_k$ and

$$\sum_{k=1}^{K} n_k = N.$$

In block k, exactly m_k units are to be assigned to treatment, so that

$$\sum_{k=1}^{K} m_k = m_1 + \cdots + m_K$$

is the total number of treated units across all blocks.

(1) Assignment mechanism for blocked randomization. Let $Z = (Z_1, \ldots, Z_N)$ denote the assignment vector, with $Z_i = 1$ if unit i is treated and $Z_i = 0$ otherwise. For each block k, define the set of units in that block as $B_k \subset \{1, \ldots, N\}$.

In a blocked randomized experiment with m_k treated units in block k, the assignment mechanism is:

$$\Pr(Z = z) = \begin{cases} \prod_{k=1}^{K} \binom{n_k}{m_k}^{-1}, & \text{if } \sum_{i \in B_k} z_i = m_k \text{ for all } k = 1, \ldots, K, \\ 0, & \text{otherwise.} \end{cases}$$

In words: within each block k, all $\binom{n_k}{m_k}$ assignments that treat exactly m_k units are equally likely, and assignments across blocks are independent. The joint probability of an assignment vector z is the product of the blockwise probabilities, provided that each block has the correct number of treated units.

(2) Reduction to complete randomization when $K = 1$. When $K = 1$, there is a single block B_1 containing all N units, so, with some abuse of notation, $n_1 = N$ and $m_1 = n_1$ is the total number of treated units (call it n_1 to match the lecture notation). The assignment mechanism becomes

$$\Pr(Z = z) = \begin{cases} \binom{N}{n_1}^{-1}, & \text{if } \sum_{i=1}^{N} z_i = n_1, \\ 0, & \text{otherwise.} \end{cases}$$

This is exactly the completely randomized design from the notes: all assignments with exactly n_1 treated units are equally likely, and any assignment with a different number of treated units has probability zero. Thus, blocked randomization reduces to complete randomization when there is only one block.

(3) Why blocking can reduce variance when potential outcomes vary across blocks. Let $\hat{\tau}_{\text{CR}}$ denote the usual difference-in-means estimator under complete randomization, and let $\hat{\tau}_{\text{BL}}$ denote the blocked difference-in-means estimator, defined as a weighted average of within-block differences:

$$\hat{\tau}_{\text{BL}} = \sum_{k=1}^{K} \frac{n_k}{N} \left(\bar{Y}_{T,k} - \bar{Y}_{C,k} \right),$$

where $\bar{Y}_{T,k}$ and $\bar{Y}_{C,k}$ are the treated and control means within block k.

Intuitively, if potential outcomes vary systematically across blocks (for example, some blocks have higher baseline outcomes than others), then complete randomization allows treated and control groups to differ in their *block composition* purely by chance. This between-block imbalance contributes to the variance of $\hat{\tau}_{\text{CR}}$.

Blocked randomization, by contrast, forces treated and control groups to be balanced within each block: in every block k, exactly m_k units are treated and $n_k - m_k$ are controls. As a result, any systematic differences in potential outcomes across blocks are *controlled by design*. The estimator $\hat{\tau}_{\text{BL}}$ compares treated and control units only within the same block, so variation due to differences in block means does not inflate its variance.

Formally, one can show that the variance of $\hat{\tau}_{\text{BL}}$ under the blocked design decomposes into a weighted sum of within-block variances, whereas the variance of $\hat{\tau}_{\text{CR}}$ under complete randomization includes both within-block and between-block components. When block means differ, the between-block component is positive, so removing it via blocking reduces the variance of the estimator.

In summary: when potential outcomes vary systematically across blocks, blocking reduces the random variation in block composition between treated and control groups, and therefore reduces the variance of the difference-in-means estimator.

Solution to Problem 5 (Other Randomized Assignment Models)

We consider three assignment mechanisms: Bernoulli trials, cluster randomization, and paired randomization. For each, we (1) describe the assignment rule, (2) state whether the total number of treated units is fixed or random, and (3) discuss one advantage and one disadvantage relative to complete randomization.

(A) Bernoulli trials with $\Pr(Z_i = 1) = p$. **(1) Assignment rule.** Each unit i is independently assigned to treatment with probability p:

$$\Pr(Z_i = 1) = p, \quad \Pr(Z_i = 0) = 1 - p,$$

and the joint assignment probability factorizes as

$$\Pr(Z = z) = \prod_{i=1}^{N} p^{z_i}(1-p)^{1-z_i}.$$

(2) Fixed or random number of treated units? The total number of treated units

$$\sum_{i=1}^{N} Z_i$$

is random. In fact,

$$\sum_{i=1}^{N} Z_i \sim \text{Binomial}(N, p).$$

(3) Advantage and disadvantage relative to complete randomization.

Advantage: The Bernoulli design is simple to implement and very flexible. Each unit can be assigned independently, which is convenient in large-scale or online settings (e.g., web experiments) where units arrive sequentially and the total sample size is not known in advance.

Disadvantage: Because the total number of treated units is random, there can be substantial imbalance in the treatment–control split, especially in smaller samples. This can increase the variance of the difference-in-means estimator relative to a completely randomized design with a fixed treated fraction.

(B) Cluster randomization. Suppose the N units are partitioned into G clusters $C_1, \ldots, C_G$, with $|C_g| = n_g$ and $\sum_{g=1}^{G} n_g = N$.

(1) Assignment rule. A simple cluster-randomized design treats entire clusters rather than individual units. For example, we may select G_1 clusters to be treated and $G_0 = G - G_1$ clusters to be control, with

$$\Pr(Z = z) = \begin{cases} \binom{G}{G_1}^{-1}, & \text{if exactly } G_1 \text{ clusters are treated,} \\ 0, & \text{otherwise.} \end{cases}$$

Within a treated cluster C_g, all units $i \in C_g$ have $Z_i = 1$; within a control cluster, all units have $Z_i = 0$.

(2) Fixed or random number of treated units? If we fix the number of treated clusters G_1, then the number of treated units

$$\sum_{i=1}^{N} Z_i = \sum_{g \in \mathcal{T}} n_g$$

is typically *not* fixed in advance unless all clusters have the same size. It is determined by which clusters are selected for treatment.

(3) Advantage and disadvantage relative to complete randomization.

Advantage: Cluster randomization is often necessary when treatment is delivered at a group level (e.g., schools, villages, firms) or when there is interference within clusters (spillovers). It respects the natural grouping of units and can be logistically simpler to implement.

Disadvantage: Because treatment is assigned at the cluster level, the effective number of independent units is the number of clusters, not the number of individuals. This typically leads to larger variances and wider confidence intervals compared to individual-level complete randomization. Inference must account for clustering (e.g., via cluster-robust standard errors).

(C) Paired randomization. Suppose units are grouped into $N/2$ pairs (assume N is even), with each pair containing two units that are similar in covariates.

(1) Assignment rule. Within each pair, exactly one unit is assigned to treatment and one to control. For pair j with units i_1 and i_2, the assignment is:

$$(Z_{i_1}, Z_{i_2}) = \begin{cases} (1,0), & \text{with probability } 1/2, \\ (0,1), & \text{with probability } 1/2. \end{cases}$$

Assignments are independent across pairs. The joint assignment probability is

$$\Pr(Z = z) = \begin{cases} 2^{-N/2}, & \text{if each pair has exactly one treated and one control unit,} \\ 0, & \text{otherwise.} \end{cases}$$

(2) Fixed or random number of treated units? The total number of treated units is fixed:

$$\sum_{i=1}^{N} Z_i = N/2.$$

(3) Advantage and disadvantage relative to complete randomization.

Advantage: Paired randomization creates very strong local balance on covariates: within each pair, treated and control units are highly similar. This can substantially reduce the variance of the difference-in-means estimator when outcomes are strongly related to the pairing covariates.

Disadvantage: Constructing good pairs requires additional design effort and good covariate information. If pairs are poorly matched, the gains in precision may be small. Analysis is also slightly more complex, as it is natural to work with pair-level differences rather than individual-level outcomes.

In all three cases, the key contrast with complete randomization is how the assignment mechanism controls (or fails to control) the number of treated units and the structure of covariate balance, which in turn affects both the variance of estimators and the practical feasibility of the design.

Solution to Problem 6 (Numerical A/B Testing Example)

We have $N = 10$ users, with 5 assigned to treatment and 5 to control. The observed outcomes (number of sessions in a week) are:

$$\text{Treatment: } 8,\ 6,\ 7,\ 9,\ 5, \qquad \text{Control: } 4,\ 6,\ 5,\ 3,\ 4.$$

Let us denote the full set of outcomes (in some fixed order) by

$$Y = (8, 6, 7, 9, 5, 4, 6, 5, 3, 4).$$

(1) Observed difference in means. The treated mean is

$$\bar{Y}_T = \frac{8+6+7+9+5}{5} = \frac{35}{5} = 7.$$

The control mean is

$$\bar{Y}_C = \frac{4+6+5+3+4}{5} = \frac{22}{5} = 4.4.$$

Therefore, the observed difference in means is

$$\hat{\tau}_{\text{obs}} = \bar{Y}_T - \bar{Y}_C = 7 - 4.4 = 2.6.$$

(2) All possible treatment assignments. The design is a completely randomized experiment with $n_1 = 5$ treated and $n_0 = 5$ control units out of $N = 10$. The set of all possible assignments is

$$\mathcal{G} = \left\{ z \in \{0,1\}^{10} : \sum_{i=1}^{10} z_i = 5 \right\},$$

and the number of such assignments is

$$|\mathcal{G}| = \binom{10}{5} = 252.$$

Under the design, each $z \in \mathcal{G}$ has equal probability

$$\Pr(Z = z) = \frac{1}{252}.$$

(3) Randomization distribution under the sharp null. Under Fisher's sharp null hypothesis of no treatment effect,

$$H_0 : Y_i(1) = Y_i(0) \quad \text{for all } i,$$

the potential outcomes are all equal to the observed outcomes, and the outcomes do not depend on Z. Thus, for any assignment $z \in \mathcal{G}$, the treated and control outcomes are just a partition of the fixed multiset

$$\{8, 6, 7, 9, 5, 4, 6, 5, 3, 4\}.$$

For a given assignment z, let $S_T(z)$ denote the sum of the outcomes in the treated group:

$$S_T(z) = \sum_{i : z_i = 1} Y_i.$$

Since there are 5 treated and 5 control units, the difference-in-means statistic can be written as

$$T(z) = \bar{Y}_T(z) - \bar{Y}_C(z) = \frac{1}{5}\sum_{i:z_i=1} Y_i - \frac{1}{5}\sum_{i:z_i=0} Y_i.$$

Let $S = \sum_{i=1}^{10} Y_i$ be the total sum of all outcomes. Here

$$S = 8 + 6 + 7 + 9 + 5 + 4 + 6 + 5 + 3 + 4 = 57.$$

Then

$$\sum_{i:z_i=0} Y_i = S - S_T(z),$$

and hence

$$T(z) = \frac{1}{5}S_T(z) - \frac{1}{5}(S - S_T(z)) = \frac{1}{5}\big(2S_T(z) - S\big).$$

Thus, the randomization distribution of $T(Z)$ under H_0 is determined by the distribution of $S_T(Z)$ when Z is uniformly distributed over all $\binom{10}{5}$ assignments.

(4) Exact p-value. The observed statistic is

$$T_{\text{obs}} = 2.6.$$

Under H_0, the randomization p-value for a two-sided test is

$$p = \Pr\big(|T(Z)| \geq |T_{\text{obs}}|\big) = \frac{\#\{z \in \mathcal{G} : |T(z)| \geq 2.6\}}{252}.$$

Using the relation

$$T(z) = \frac{1}{5}(2S_T(z) - 57),$$

the condition $|T(z)| \geq 2.6$ is equivalent to

$$|2S_T(z) - 57| \geq 13 \quad \Longleftrightarrow \quad 2S_T(z) - 57 \geq 13 \;\text{ or }\; 2S_T(z) - 57 \leq -13,$$

that is,

$$S_T(z) \geq 35 \quad \text{or} \quad S_T(z) \leq 22.$$

A careful enumeration of all $\binom{10}{5} = 252$ subsets of size 5 from the multiset $\{8, 6, 7, 9, 5, 4, 6, 5, 3, 4\}$ shows:

- There are 1 subset with sum $S_T = 21$, - 4 subsets with sum $S_T = 22$, - 4 subsets with sum $S_T = 35$, - 2 subsets with sum $S_T = 36$,

and no subsets with sums below 21 or above 36. Therefore, the number of assignments with $S_T \leq 22$ or $S_T \geq 35$ is

$$1 + 4 + 4 + 2 = 11.$$

Hence the exact two-sided p-value is

$$p = \frac{11}{252} \approx 0.0437.$$

The observed assignment (with treated outcomes $8, 6, 7, 9, 5$) has

$$S_T = 8 + 6 + 7 + 9 + 5 = 35,$$

so $T_{\text{obs}} = 2.6$ lies in the extreme tail of the randomization distribution, and the corresponding p-value is about 4.4%.

Solution to Problem 7 (Finite Population Perspective)

We consider a randomized experiment with fixed potential outcomes

$$\{Y_i(1), Y_i(0)\}_{i=1}^N.$$

(1) Why is the only source of randomness the treatment assignment? In the Neymanian (design-based) framework, the potential outcomes

$$Y_i(1),\ Y_i(0)$$

are treated as fixed, unknown numbers for each unit i. They do not have a probability distribution; they are not random variables. The only random object is the assignment vector

$$Z = (Z_1, \ldots, Z_N),$$

which is generated according to a known assignment mechanism (e.g., complete randomization).

The observed outcome for unit i is

$$Y_i^{\text{obs}} = Z_i Y_i(1) + (1 - Z_i) Y_i(0),$$

which is a deterministic function of $(Y_i(1), Y_i(0), Z_i)$. Since the potential outcomes are fixed, all randomness in Y_i^{obs} comes from the randomness in Z_i. Therefore, in this framework, *the only source of randomness is the treatment assignment.*

(2) The sample average treatment effect is fixed. The finite-population average treatment effect (PATE) is defined as

$$\tau = \frac{1}{N} \sum_{i=1}^{N} \big(Y_i(1) - Y_i(0)\big).$$

Each term $Y_i(1) - Y_i(0)$ is a fixed number, and the sum is taken over a fixed finite set of units. Therefore, τ is itself a fixed (non-random) quantity. It does not depend on the realized assignment Z; it is defined purely in terms of the science (the collection of potential outcomes).

In particular, τ is not a parameter of a probability distribution in this framework; it is a finite-population quantity that we aim to estimate using the randomness induced by Z.

(3) Contrast with a superpopulation model. In a superpopulation model, we imagine that the units in the experiment are drawn from an infinite (or very large) population according to some probability law. The potential outcomes $Y_i(1)$ and $Y_i(0)$ are then viewed as random draws from a joint distribution, and the parameter of interest is often the *superpopulation* average treatment effect

$$\tau^* = \mathbb{E}\big[Y(1) - Y(0)\big],$$

where the expectation is taken over the superpopulation distribution.

In that setting, randomness comes from two sources:

1. The sampling of units from the superpopulation. 2. The random assignment of treatment to sampled units.

By contrast, in the Neymanian finite-population perspective:

- The set of N units and their potential outcomes are fixed. - There is no superpopulation distribution. - The only randomness is the treatment assignment.

Thus, the Neymanian framework conditions on the realized finite population and uses the randomization mechanism to justify probability statements, whereas the superpopulation model introduces an additional layer of randomness through sampling from a hypothetical population.

Solution to Problem 8 (Randomization-Based vs Model-Based Inference)

We compare a randomization-based t-test (using the known assignment mechanism), and a linear regression with homoskedastic errors,

applied to data from a completely randomized experiment.

(1) Why do these procedures often yield similar numerical results? In a completely randomized experiment with a large sample and reasonably well-behaved outcomes, the difference-in-means estimator

$$\hat{\tau} = \bar{Y}_T - \bar{Y}_C$$

is approximately Normaly distributed under the randomization distribution (by a finite-population CLT), and its variance is well estimated by Neyman's conservative variance estimator.

A linear regression of Y on Z (and possibly covariates) with homoskedastic errors also produces an estimate of the treatment effect and a standard error. When the regression model is correctly specified (or at least not badly misspecified) and errors are close to homoskedastic, the regression-based standard error is numerically close to the Neyman variance-based standard error.

Thus, in many practical randomized experiments, both approaches are essentially analyzing the same contrast (the difference in means) and using similar variance formulas, leading to similar p-values and confidence intervals.

(2) A setting where regression-based standard errors may be misleading. Regression-based standard errors rely on modeling assumptions, such as:

- homoskedasticity of errors, and
- correct functional form (e.g., linearity in covariates).

In a randomized experiment with highly heteroskedastic outcomes (e.g., heavy-tailed or skewed distributions, or variances that differ sharply between treated and control groups), the usual homoskedastic regression standard errors can be severely biased. They may understate or overstate the true variability of the estimator.

By contrast, randomization-based inference (or Neyman's design-based variance) remains valid because it does not assume any particular outcome distribution; it uses only the known assignment mechanism.

(3) Why Paul Rosenbaum argues that randomization-based inference is conceptually primary. Paul Rosenbaum's argument is that in a randomized experiment:

- The *only* known and controlled source of randomness is the assignment mechanism.
- Potential outcomes are fixed attributes of units.
- Probability statements should therefore be derived from the randomization distribution, not from an assumed model for $Y \mid X, Z$.

Randomization-based inference:

- makes explicit what is assumed (the design) and what is not (the outcome model),
- yields exact or finite-sample-valid tests under sharp nulls,
- does not rely on unverifiable distributional assumptions about outcomes.

Model-based (regression) inference can be a convenient approximation, and often works well, but it is conceptually secondary: its validity depends on assumptions about the outcome distribution that are not guaranteed by the design. Randomization-based inference, in contrast, derives its validity directly from the physical act of randomization, which is known and under the experimenter's control.